

required supplied structure $\rightarrow$ reconstructed quantity

R0

order only

$\rightarrow$ flat-sprinkling dimension; raw chain statistics

bounded by: model assumptions; no metric scale

R1

order + global density

$\rightarrow$ timelike proper time

bounded by: needs a density

R2

order + observer clock

$\rightarrow$ radar time; unsigned distance

bounded by: sign undetermined

R3

order + observer + orientation

$\rightarrow$ signed coords; Lorentz map

bounded by: calibrated separation

R4

order + observer + oriented atlas

$\rightarrow$ Poincaré transition maps

bounded by: calibrated charts

R5

order + global/local measure

$\rightarrow$ volume; coarse-grain stability

bounded by: global/local measure supplied